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RAZOR ASSEMBLY

Abstract

A razor assembly is proposed. The assembly may include a razor cartridge including a blade housing configured to longitudinally accommodate at least one razor blade each having a cutting edge. The razor cartridge may include a cartridge top surface where the cutting edge is exposed and a cartridge bottom surface opposite to the cartridge top surface. The assembly may also include a razor handle connected to the razor cartridge, the razor handle including a light source unit configured to emit light from one side of the razor handle. The light emitted from the light source unit may be reflected by a reflective surface included in the razor cartridge and may be emitted to the outside at the cartridge top surface.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION
[0001] This application claims priority to Korean Patent Application No. 10-2024-0025992, filed in Korea on Feb. 22, 2024, which is incorporated herein by reference in its entirety.
BACKGROUND
Technical Field
[0002] The present disclosure relates to a razor assembly.
Description of Related Technology
[0003] Generally, a razor assembly, which is known as a wet razor, includes a razor cartridge and a razor handle. Recently, research is being actively conducted on ways to improve shaving performance by assisting the function of a razor blade along with the razor blade included in the cartridge.
SUMMARY
[0004] One aspect is a razor assembly that may provide a user with an extensive range of clear visual information about the skin area to be shaved before shaving begins.
[0005] Another aspect is a razor assembly, comprising: a razor cartridge comprising a blade housing configured to longitudinally accommodate at least one razor blade each having a cutting edge, the razor cartridge comprising a cartridge top surface where the cutting edge is exposed and a cartridge bottom surface opposite the cartridge top surface; and a razor handle connected to the razor cartridge, the razor handle comprising a light source unit configured to emit light from one side of the razor handle, wherein the light emitted from the light source unit is reflected by a reflective surface included in the razor cartridge and is emitted to the outside at the cartridge top surface.
[0006] As described above, according to an embodiment of the present disclosure, there is a benefit of providing a user with an extensive range of clear visual information about the skin area to be shaved before shaving begins.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1** is an exploded perspective view of a razor assembly according to an embodiment of the present disclosure.
[0008] FIG. **2** is a coupled perspective view of the razor assembly according to an embodiment of the present disclosure.
[0009] FIG. **3** is a side cross-sectional view of the razor assembly of FIG. **2**, taken in a direction of A-A′.
[0010] FIG. **4** is an enlarged view of part B in the side cross-sectional view of FIG. **3**.
[0011] FIG. **5** is a diagram illustrating a light source unit and light emitting area in the razor assembly of FIG. **2** with a portion of a connecting head omitted.
[0012] FIG. **6** is a diagram illustrating internal configurations of a head adapter and the connecting head in the razor assembly of FIG. **2** with a portion of the head adapter and the connecting head omitted.
DETAILED DESCRIPTION
[0013] A user of a wet razor may hold the razor handle in his/her hand and cut body hair by contacting the razor cartridge to the skin and stroking the same.

[0014] In this case, since the razor blade included in the cartridge comes into direct contact with the skin, irritation is applied to the skin where the body hair to be cut is positioned. Accordingly, a user may be required to perform skilled work, such as not shaving a specific area or changing a shaving direction, depending on the condition of a skin area subject to shaving.
[0015] However, due to the structure of the razor assembly, it is difficult for a user to know at first hand the part where the razor blade will contact the skin during shaving, so it is very difficult for the user to avoid shaving certain areas of the skin or to perform shaving in a different way.
[0016] In particular, when a user shaves a part where the skin is highly curved and a shadow is formed, such as under the chin, clear visual information about the part needs to be provided. For example, when the user is able to visually check the sensitive condition of the skin before contacting the razor cartridge to the skin, skin irritation may be prevented by assisting the user in determining how much and whether to press the razor cartridge against the skin.
[0017] In addition, when precise shaving is required, such as shaving small areas such as under the nose and around the edges of the lips, or trimming sideburns, it is important to allow a user to see exactly which part of the skin the razor blade will be shaving.
[0018] Accordingly, there is a need for a razor assembly that may provide a user with an extensive range of clear visual information about the skin area to be shaved before shaving begins.
[0019] Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. It is to be noted that in giving reference numerals to components of each of the accompanying drawings, the same components will be denoted by the same reference numerals even though they are illustrated in different drawings. Further, in describing exemplary embodiments of the present invention, well-known functions or constructions will not be described in detail since they may unnecessarily obscure the understanding of the present invention.
[0020] Terms ‘first’, ‘second’, i), ii), a), b), and the like, will be used in describing components according to embodiments of the present disclosure. These terms are only for distinguishing the components from other components, and the nature, sequence, order, or the like of the components are not limited by the terms. Throughout the present specification, unless explicitly described to the contrary, “including” or “comprising” any components will be understood to imply the inclusion of other elements rather than the exclusion of any other elements.
[0021] FIG. **1** is an exploded perspective view of a razor assembly **10** according to an embodiment of the present disclosure.
[0022] FIG. **2** is a coupled perspective view of the razor assembly **10** according to an embodiment of the present disclosure.
[0023] FIG. **3** is a side cross-sectional view of the razor assembly **10** of FIG. **2**, taken in a direction of A-A′.
[0024] FIG. **4** is an enlarged view of part B in the side cross-sectional view of FIG. **3**.
[0025] FIG. **5** is a diagram illustrating a light source unit **201** and light emitting area **350** in the razor assembly **10** of FIG. **2** with a portion of a connecting head **242** and **244** omitted.
[0026] Referring to FIGS. **1** to **5**, the razor assembly **10** according to an embodiment of the present disclosure includes all or part of a razor cartridge **100**, a connector **160**, and a razor handle **200**.
[0027] The razor cartridge **100** includes a blade housing **120** configured to longitudinally accommodate at least one razor blade **140**, each having a cutting edge (not shown). Herein, the longitudinal direction may refer to a length direction of the razor cartridge **100**. For example, the longitudinal direction may be parallel to the X-axis based on FIGS. **1**, **2**, and **5**.
[0028] The surface where the cutting edge is exposed on the razor cartridge **100** may be defined as a cartridge top surface **102**, and the surface opposite to the cartridge top surface **102** may be defined as a cartridge bottom surface **104**.
[0029] The connector **160** may be coupled to one side of the blade housing **120**. For example, the connector **160** may be manufactured as a separate member from the blade housing **120** and then

mounted on the blade housing **120**. In this case, the connector **160** may be fixed to the blade housing **120** or may be pivotally connected to the blade housing **120**.

[0030] Alternatively, the connector **160** may be formed to extend from one side of the blade housing **120** and be manufactured integrally with the blade housing **120**. In this case, the connector **160** may be formed by injecting from a bottom surface **104** of the blade housing **120**, but is not necessarily limited thereto.

[0031] In addition, the connector **160** may be disposed on at least a portion of the bottom surface **104** of the blade housing **120**, excluding the area where the razor blade **140** is disposed. Preferably, the connector **160** may be disposed at front of the razor blade **140** with respect to a shaving direction, but is not necessarily limited thereto. The connector **160** may be disposed on at least a portion of the area where the razor blade **140** is disposed on the bottom surface **104** of the blade housing **120**, and may be disposed at rear of the razor blade **140** with respect to the shaving direction.

[0032] The razor handle **200** includes a light source unit **201** connected to the razor cartridge **100** and configured to emit light from one side of the razor handle **200**. The light emitted from the light source unit **201** may be reflected by a reflective surface **300** included in the razor cartridge **100** and emitted to the outside at the cartridge top surface **102**.

[0033] In this case, the reflective surface **300** may be positioned inside the razor cartridge **100** and not exposed to the outside, and accordingly, the light incident on the reflective surface **300** and the light reflected and emitted from the reflective surface **300** may not be emitted to the outside until reaching the cartridge top surface **102**.

[0034] In addition, in order to prevent the light from being emitted to the outside until reaching the cartridge top surface **102**, the space through which light may pass from the light source unit **201** to the reflective surface **300** and from the reflective surface **300** to the cartridge top surface **102** may be maintained in a sealed state without cracking.

[0035] For proper reflection, the reflective surface **300** may be provided with a reflective member such as a mirror, or the surface of the reflective surface **300** may be coated or a special pattern may be formed. However, an embodiment of the present disclosure is not necessarily limited thereto.

[0036] Referring to FIG. **4**, the reflective surface **300** is illustrated as a plane perpendicular to a height direction perpendicular to the longitudinal direction, for example, a direction parallel to the Z-axis of FIG. **4**, but is not necessarily limited thereto. For example, the reflective surface **300** may be configured of a curved surface, or may be a surface including both a curved surface and a flat surface.

[0037] In the drawing, the razor cartridge **100** is illustrated as including one reflective surface **300**, but the reflective surface **300** does not necessarily have to be configured of one, and may be configured of a plurality of reflective surfaces **300** depending on structural factors of the razor assembly **10**. When the razor cartridge **100** includes the plurality of reflective surfaces, light emitted from the light source unit **201** may be reflected by the number of the plurality of reflective surfaces and be emitted to the outside at the cartridge top surface **102**.

[0038] In addition, when the channel extending from one side of the light source unit **201** to the reflective surface **300** is referred to as a first light channel **400**, and the channel extending from the reflective surface **300** to the cartridge top surface **102** is referred to as a second light channel **450**, the light emitted from the light source unit may move along the first light channel **400** and the second light channel **450**.

[0039] In FIG. **4**, it is illustrated that the reflective surface **300** is configured of one and there are two light channels **400** and **450**. However, when the reflective surface **300** is configured in a plural number, additional light channels may be disposed between the first light channel **400** and the second light channel **450**.

[0040] In addition, the center of the first light channel **400** and the center of the second light channel **450** may not be positioned on the same line. When viewed from one longitudinal side of the razor cartridge **100**, an extension direction of the first light channel **400** and an extension direction of the second light channel **450** may not be the same.

[0041] The razor assembly **10** according to an embodiment of the present disclosure is configured so that the light emitted from the light source unit **201** is reflected on the reflective surface **300** of the razor cartridge **100** and emitted to the outside of the razor cartridge **100**. The light emitted from the light source unit **201** does not necessarily have to be directed directly toward the cartridge top surface **102**.

[0042] Accordingly, by including the reflective surface **300** in the razor cartridge **100**, the razor assembly **10** according to an embodiment of the present disclosure may have a greater degree of freedom in design in disposing the light source unit **201**.

[0043] In addition, light emitted to the outside from the cartridge top surface **102** may provide high visibility to a user when touching the skin of the user. Accordingly, the user may clearly recognize the skin area to be shaved before shaving begins.

[0044] The light touching the skin of a user not only provides the user with visual information about the area to be shaved, but in some cases may also provide various cosmetic effects to the skin of the user.

[0045] The razor cartridge **100** may include the light emitting area **350** disposed at front of the at least one razor blade **140** with respect to the shaving direction. In this case, the light emitted from the light source unit **201** may be emitted to the outside through the light emitting area **350**.

[0046] For example, referring to FIG. **5**, the light emitting area **350** may be disposed ahead of the frontmost razor blade with respect to the shaving direction of the at least one razor blade **140**, thereby allowing a user to receive clearer visual information about the area to be shaved before shaving begins.

[0047] The light emitting area **350** may include a transparent material to better emit light, and the intensity of the emitted light may be adjusted depending on the degree of transparency of the light emitting area **350**. The transparent material includes a component that may remove dead skin cells, so the light emitting area **350** may perform the function of removing dead skin cells.

[0048] However, the light emitting area **350** does not necessarily include a transparent material so that light is emitted through the transparent material, and the light emitting area **350** is an area open in the height direction without a transparent material and may be configured to emit light reflected by the reflective surface **300**.

[0049] The light emitted from the light source unit **201** may be configured to be reflected by the reflective surface **300** and be emitted from the entire area of the light emitting area **350**, but is not necessarily limited to this, and may be configured to be emitted only from a portion of the light emitting area **350**. In this case, the light emitting area **350** may include an area where light is emitted and an area where light is not emitted.

[0050] In addition, the razor handle **200** may further include a cover unit **280**, at least portion of which is disposed at front of the light source unit **201** with respect to a direction in which the light is emitted from the light source unit. Referring to FIG. **4**, the light emitted from the light source unit **201** may be diffused from the light source unit **201** and then aligned in a direction parallel to or convergent to any direction through the cover unit **280**. Herein, convergence includes not only cases where light is concentrated toward a certain focus, but also cases where light diverges but the degree of divergence becomes smaller compared to the initial incident direction. The light passing through the cover unit **280** may be reflected by the reflective surface **300** and then diffused and emitted through the light emitting area **350**. In FIG. **4**, the light passing through the cover unit **280** and the light reflected on the reflective surface **300** are shown with a single arrow. However, this merely shows a representative light path by way of example, and the light passing through the cover unit **280** and the light reflected on the reflective surface **300** may appear as a plurality of light paths that are parallel or converge in one direction.

[0051] Since the light emitted from the light source unit **201** may be aligned in a direction parallel to or convergent to any one direction while passing through the first light channel **400** and the second light channel **450**, the light transmission efficiency up to the light emitting area **350** may be increased.

[0052] A pattern may be formed on at least one surface of the light emitting area **350**, and light passing through the light emitting area **350** may be emitted in various directions depending on the pattern formed. For example, in order to improve shaving efficiency, the surface pattern of the light emitting area **350** may be configured to narrowly focus light in a specific area or spread widely, and depending on the characteristics of shaving, the surface pattern may be uniform or of various types.

[0053] Since the light aligned in one direction by the cover unit **280** may spread through the light emitting area **350**, the light may be emitted to a wider range of skin areas. For example, referring again to FIG. **5**, a longitudinal width of the second light channel **450** on a side adjacent to the cartridge top surface **102** may be greater than a longitudinal width of the first light channel **400** on a side adjacent to the light source unit **201**.

[0054] However, the light emitted from the light source unit **201** does not necessarily have to be diffused by the light emitting area **350**, and it is also possible to diffuse the light by adjusting the degree of curvature of the reflective surface **300**. For example, the reflective surface **300** may be configured to be flat, but may also be configured to be curved to form a predetermined angle, or may be configured to be convex upward or downward.

[0055] The light source unit **201** included in the razor handle **200** may include a first circuit board **201_2** including a support surface, and a light emitting unit **201_4** disposed on the support surface and electrically connected to the first circuit board **201_2**.

[0056] For example, the first circuit board **201_2** may be relatively long in the longitudinal direction, and the light emitting unit **201_4** may include a plurality of LEDs (Light Emitting Diodes), but is not limited thereto.

[0057] Additionally, the razor handle **200** may include a grip portion **207**, a connecting portion **202**, a bottom cap **208**, and a head portion **240**.

[0058] The grip portion **207** may include a second circuit board **203** that is electrically connected to a power supply unit **204**.

[0059] Herein, a switch unit **203_5** may be disposed on the second circuit board **203**. A button unit **206** may be disposed on one side of the grip portion **207** so that the power supply unit **204** may supply power when a user presses the button unit **206**. However, the power supply method of the power supply unit **204** is not necessarily limited to the aforementioned method.

[0060] The position of the button unit **206** is not necessarily limited to the position illustrated in FIGS. **1** to **3**, and may be positioned at one end of the grip portion **207** adjacent to the bottom cap **208**, and may be positioned at a bottom surface of the bottom cap **208**.

[0061] In addition, when the button unit **206** is pressed again after power is supplied, the power supply may be cut off, or the power supply may be configured to be automatically cut off after a preset time has elapsed after power has been supplied.

[0062] The connecting portion **202** may electrically connect the second circuit board **203** and the light source unit **201**. In this case, the power supplied from the power portion **204** may be transmitted to the light emitting unit **201_4** through the second circuit board **203**, the connecting portion **202**, and the first circuit board **201_2** in that order. The connecting portion **202** is preferably a circuit board configured of at least a portion of a flexible material in order to secure structural design freedom for other components, but is not necessarily limited thereto.

[0063] At least a portion of the power supply unit **204**, the second circuit board **203**, and the connecting portion **202** may be protected by being accommodated in an accommodation housing **205** disposed within the grip portion **207**, but is not necessarily limited thereto.

[0064] The bottom cap **208** may be detachably connected to one end of the grip portion **207**. When the life of the power supply unit **204** has expired, a user may replace the configuration of the power supply unit **204** by separating the bottom cap **208** from the grip portion **207** in a direction parallel to the extension direction of the grip portion **207**. However, the power supply unit **204** does not necessarily have to be configured to be replaceable, and the power supply unit **204** may be configured to be charged in a wired or wireless manner while accommodated in the grip portion **207**.

[0065] The head portion **240** may accommodate at least a portion of the connecting portion **202**, one side of which may be connected to the grip portion **207** and the other side of which may be connected to the razor cartridge **100**. In this case, the light source unit **201** may be positioned on one side of the head portion **240**, and more specifically, at least a portion of the light source unit **201** is supported by the head portion **240** so that the position of the light source unit **201** may be fixed to the head portion **240**.

[0066] Hereinafter, with further reference to FIG. **6**, technical features of the razor assembly **10** according to an embodiment of the present disclosure will be described in detail.

[0067] FIG. **6** is a diagram illustrating internal configurations of a head adapters **246** and **248** and the connecting heads **242** and **244** in the razor assembly **10** of FIG. **2** with a portion of the head adapters **246** and **248** and the connecting heads **242** and **244** omitted.

[0068] Referring to FIGS. **1** to **6**, the razor handle **200** may further include a first restoring force providing portion **210**, a coupling unit **230**, an ejecting unit **220**, and a second restoring force providing portion **250**.

[0069] In addition, at least a portion of each of the first restoring force providing portion **210**, the ejecting unit **220**, the coupling unit **230**, and the second restoring force providing portion **250** may be included within the head portion **240**. More specifically, a portion of the first restoring force providing portion **210** may protrude from one side of the head portion **240**, and the ejecting unit **220**, the coupling unit **230**, and the second restoring force providing portion **250** may be disposed inside the head portion **240**.

[0070] The razor cartridge **100** is pivotable about a first pivot axis ax**1** parallel to the longitudinal direction with respect to the razor handle **200**.

[0071] In this case, the first restoring force providing portion **210** may be configured to provide a restoring force used to restore the razor cartridge **100** to a first neutral position when the razor cartridge **100** pivots around the first pivot axis ax**1**. Herein, the first neutral position refers to a position in which the razor cartridge **100** is not pivoted about the first pivot axis ax**1**.

[0072] In the first neutral position, the razor cartridge may be pivoted in the range of 20° to 60°, and preferably in the range of 32° to 50°, with respect to the direction parallel to the Y-axis direction of FIG. **3**.

[0073] Referring to FIG. **4**, the first pivot axis ax**1** may be disposed between the light source unit **201** and the at least one razor blade **140**.

[0074] In the case where the light emitted from the light source unit **201** is not reflected but is directly emitted to the outside of the razor cartridge **100**, the light source unit **201** may be disposed adjacent to the rear of the light emitting area **350** with respect to the direction in which the light is emitted.

[0075] The razor assembly **10** according to an embodiment of the present disclosure is configured to indirectly emit light emitted from the light source unit **201** to the outside of the razor cartridge **100** using the reflective surface **300**, the position of the light source unit **201** may be freely set. Accordingly, with respect to the shaving direction, the light source unit **201** may be disposed ahead of the first pivot axis ax**1**, and the first pivot axis ax**1** may be positioned close to the front of the at least one razor blade **140**. Accordingly, a smoother pivoting feeling of the razor cartridge **100** may be provided.

[0076] Referring to FIGS. **3** and **4**, when the razor cartridge **100** pivots around the first pivot axis ax**1**, the angle formed by the reflective surface **300** with the first circuit board **201_2** varies, and accordingly, the amount of light reaching the cartridge top surface **102** may also vary.

[0077] A user generally brings the razor cartridge **100** close to the skin in front of a mirror to perform shaving. In this case, the user uses the light emitted from the razor cartridge **100** to check the shaving area of the at least one razor blade **140**. In this case, the cartridge top surface **102** is spaced apart from the skin, and in this state, the razor cartridge **100** is in the first neutral position. Accordingly, in order to improve the perception of a user of the position of the shaving area, there is a need for the largest amount of light to be emitted from the razor cartridge **100** when in the first neutral position. Accordingly, the amount of light emitted to the outside from the cartridge top surface **102** may be greatest in the first neutral position.

[0078] In this case, in the first neutral position, the first circuit board **201_2** may be perpendicular to the cartridge top surface **102**, and accordingly, when the light emitted from the light source unit **201** is reflected on the reflective surface **300**, a relatively large amount of light may reach the cartridge top surface **102**.

[0079] The first restoring force providing portion **210** may include a first plunger **212** and a first elastic member **214**. In this case, the first plunger **212** is configured to transmit a restoring force through a cam action with the razor cartridge **100**, and the first elastic member **214** is configured to provide an elastic force to the first plunger **212**. Herein, when the direction in which the first elastic member **214** and the first plunger **212** are coupled is referred to as a coupling direction, the coupling direction may mean the positive Y-axis direction in FIGS. **1** to **6**.

[0080] For example, when the razor cartridge **100** pivots about the first pivot axis ax**1**, the first plunger **212** may be configured to move in the Y-axis direction of FIGS. **1** to **6** according to pivot about the first pivot axis ax**1** of the razor cartridge **100**. In addition, the first elastic member **214** may be configured to contract when the first plunger **212** moves in a negative Y-axis direction. In this case, the first elastic member **214** may provide a restoring force in a positive Y-axis direction to the first plunger **212**.

[0081] The coupling unit **230** may be configured to be coupled to the connector **160**, and in this case, the first pivot axis ax**1** may pass through one end of the coupling unit **230** in a coupling direction or may be in contact with the end. For example, the coupling unit **230** may be configured to include two arms and two elastic members, but is not necessarily limited thereto.

[0082] The ejecting unit **220** is configured to separate the coupling unit **230** from the connector **160**. For example, in the case where the coupling unit **230** includes two arms, when moving in a coupling direction, the ejecting unit **220** may be configured to narrow the longitudinal distance between the two arms or to elastically deform the two arms. The coupling unit **230** may be separated from the connector **160** by narrowing the longitudinal distance between the two arms or elastically deforming the two arms.

[0083] The movement of the ejecting unit **220** in the coupling direction may be possible by a user moving an operating unit **270** coupled to the ejecting unit **220** on the outside of the head portion **240** in the positive Y-axis direction.

[0084] In addition, one end of the first elastic member **214** is connected to the rear end of the first plunger **212** with respect to the coupling direction, and the other end is connected to the ejecting unit **220**, so that when the ejecting unit **220** moves in the coupling direction, the first elastic member **214** may provide a restoring force to the ejecting unit **220**.

[0085] Although not shown in the drawing, when the razor cartridge **100** pivots about the first pivot axis ax**1**, it is also possible to configure the power supply unit **204** and the switch unit **203_5** to be spaced apart by the first restoring force providing portion **210** and/or the ejecting unit **220**, or to block power transmission to the connecting portion **202**. In this case, unnecessary power consumption during the shaving process may be reduced.

[0086] The head portion **240** may include the connecting head **242** and **244** and the head adapters **246** and **248**. One side of the connecting heads **242** and **244** may be connected to the razor cartridge. The head adapters **246** and **248** may be connected to one side of the grip portion **207** and are configured to surround at least a portion of the connecting heads **242** and **244**. In this case, at least a portion of the light source unit **201** may be supported by one side of the connecting heads

242 and **244** that are relatively more adjacent to the razor cartridge **100**.
[0087] In FIG. **1**, the connecting heads **242** and **244** are illustrated as including an upper head **242** and a lower head **244**, and the head adapters **246** and **248** are illustrated as including an upper adapter **246** and a lower adapter **248**, but is not necessarily limited thereto. The connecting heads **242** and **244** and/or the head adapters **246** and **248** may be configured in a single form in which the upper and lower portions of each are coupled.
[0088] The connecting heads **242** and **244** may be pivotable about a second pivot axis ax**2**, which is a direction perpendicular to the longitudinal direction with respect to the head adapters **246** and **248**. In other words, the razor assembly **10** according to an embodiment of the present disclosure may provide not only pivot about the first pivot axis ax**1** but also pivot around the second pivot axis ax**2**. Herein, the second pivot axis may refer to a direction parallel to the Z axis in, for example, FIGS. **1** to **6**.
[0089] In order to pivot the connecting heads **242** and **244** about the second pivot axis ax**2** with respect to the head adapters **246** and **248**, the head adapters **246** and **248** and the connecting heads **242** and **244** may be fastened by at least one fastening member **260**, in which case the at least one fastening member **260** may penetrate the head adapters **246** and **248**, the connecting portion **202**, and the connecting heads **242** and **244**.
[0090] In order for the at least one fastening member **260** to penetrate the connecting portion **202**, the connecting portion **202** may include a through hole **202_5**.
[0091] In this case, when the connecting heads **242** and **244** pivot about the second pivot axis ax**2**, the second restoring force provider **250** is configured to provide a restoring force to restore the connecting heads **242** and **244** to the second neutral position. Herein, the second neutral position means a position in which the connecting heads **242** and **244** are not pivoted about the second pivot axis ax**2**.
[0092] For the design freedom of the head portion **240** and the setting of the light emitting direction of the light source unit **201**, at least a part of the connecting portion **202** may include a first bending area **302** and a second bending area **301**.
[0093] For example, referring to FIG. **4**, when viewed from one side in the longitudinal direction, the first bending area **302** may be formed to be convex downward, and the second bending area **301** may be formed to be convex upward.
[0094] At least a portion of the connecting heads **242** and **244** may be formed to be convex downward when viewed from one longitudinal side in order to couple with the head adapters **246** and **248** or accommodate the internal components of the connecting heads **242** and **244**. In this case, in the second neutral position, at least a portion of the connecting heads **242** and **244** may be positioned at an upper portion of the first bending area **302** with respect to a direction parallel to the second pivot axis ax**2**. As the connecting portion **202** includes the first bending area **302**, the design freedom of the connecting heads **242** and **244** may be increased.
[0095] In addition, when the connecting heads **242** and **244** pivot about the second pivot axis ax**2** with respect to the head adapters **246** and **248**, the connecting portion **202** may be freely folded or unfolded using the first bending area **302** and/or the second bending area **301**. As a result, the electrical connection between the first circuit board **201_2** and the second circuit board **203** may be smoothly maintained regardless of the degree of pivot of the connecting heads **242** and **244** about the second pivot axis ax**2**.
[0096] However, the connecting portion **202** does not necessarily have to be configured to fold or unfold according to pivot about the second pivot axis ax**2** of the connecting heads **242** and **244**. It is also possible to dispose the connecting portion **202** so as not to interfere with the pivot of the connecting heads **242** and **244** about the second pivot axis ax**2**.
[0097] The connecting heads **242** and **244** may be configured to support at least a portion of the light source unit **201** so that the light emitted from the light source unit **201** is directed toward the reflective surface **300**. In this case, at least a portion of the connecting heads **242** and **244** may be

configured to support the connecting portion **202** at a lower portion of the second bending area **301** with respect to a direction parallel to the second pivot axis ax**2**. Since the connecting part **202** includes the second bending area **301**, the light emitted from the light source unit **201** may be directed to the reflective surface **300**.

[0098] In addition, the connecting portion **202** may be configured to connect the first bending area **302** and the second bending area **301** to each other. For example, the second bending area **301** may be formed immediately after the first bending area **302** is formed with respect to the coupling direction. Accordingly, the support of the connecting heads **242** and **244** with respect to the connecting portion **202** may be made more stable, and the connecting portion **202** may be prevented from moving in the height direction relative to the connecting heads **242** and **244** and/or the head adapters **246** and **248**.

[0099] The second restoring force providing portion **250** may include a second plunger **252** and a second elastic member **254**. In this case, the second plunger **252** is configured to transmit a restoring force through a cam action with the connecting heads **242** and **244**, and the second elastic member **254** provides an elastic force to the second plunger **252**.

[0100] For example, when the connecting heads **242** and **244** pivot about the second pivot axis ax**2**, the second plunger **252** may be configured to move in the Y-axis direction of FIGS. **1** to **6** according to pivot about the second pivot axis ax**2** of the razor cartridge **100**. In addition, the second elastic member **254** may be configured to contract when the second plunger **252** moves in a negative Y-axis direction. In this case, the second elastic member **254** may provide a restoring force in a positive Y-axis direction to the second plunger **252**.

[0101] Although not shown in the drawing, when the connecting heads **242** and **244** pivot around the second pivot axis ax**2**, it is also possible to configure the power supply unit **204** and the switch unit **203_5** to be spaced apart by the second restoring force providing portion **250**, or to block power transmission to the connecting portion **202**. In this case, unnecessary power consumption during the shaving process may be reduced.

[0102] The spirit of the present embodiment is illustratively described hereinabove. It will be appreciated by those skilled in the art to which the present embodiment pertains that various modifications and alterations may be made without departing from the essential characteristics of the present embodiment. Accordingly, the present embodiments are not to limit the spirit of the present embodiment, but are to describe the spirit of the present embodiment. The technical idea of the present embodiment is not limited to these embodiments. The scope of the present embodiment should be interpreted by the following claims, and it should be interpreted that all the spirits equivalent to the following claims fall within the scope of the present embodiment.

Claims

1. A razor assembly, comprising: a razor cartridge comprising a blade housing configured to longitudinally accommodate at least one razor blade each having a cutting edge, the razor cartridge comprising: a cartridge top surface where the cutting edge is exposed, and a cartridge bottom surface opposing the cartridge top surface; and a razor handle connected to the razor cartridge, the razor handle comprising a light source unit configured to emit light from one side of the razor handle, wherein the light emitted from the light source unit is configured to be reflected by a reflective surface included in the razor cartridge and is configured to be emitted to the outside at the cartridge top surface.

2. The razor assembly of claim 1, wherein: the razor cartridge comprises a light emitting area disposed at front of the at least one razor blade with respect to a shaving direction; and the light emitted from the light source unit is configured to be emitted to the outside through the light emitting area.

3. The razor assembly of claim 2, wherein the light emitting area comprises a transparent material.

4. The razor assembly of claim 2, further comprising a cover unit, at least portion of which is disposed at a front of the light source unit with respect to a direction in which the light is emitted from the light source unit, wherein: the light emitted from the light source unit is configured to be diffused from the light source unit and then aligned in a direction parallel to or convergent to any one direction through the cover unit; and the light passing through the cover unit is configured to be reflected by the reflective surface and then diffused and emitted through the light emitting area according to a pattern formed on at least one surface of the light emitting area.
5. The razor assembly of claim 1, wherein the reflective surface is positioned inside the razor cartridge.
6. The razor assembly of claim 1, further comprising: a first light channel extending from one side of the light source unit to the reflective surface; and a second light channel extending from the reflective surface to the cartridge top surface, wherein: the light emitted from the light source unit is configured to move along the first light channel and the second light channel; and a longitudinal width of the second light channel on a side adjacent to the cartridge top surface is greater than a longitudinal width of the first light channel on a side adjacent to the light source unit.
7. The razor assembly of claim 1, further comprising: a first light channel extending from one side of the light source unit to the reflective surface; and a second light channel extending from the reflective surface to the cartridge top surface, wherein the center of the first light channel and the center of the second light channel are not positioned on the same line.
8. The razor assembly of claim 1, further comprising: a first light channel extending from one side of the light source unit to the reflective surface; and a second light channel extending from the reflective surface to the cartridge top surface, wherein, when viewed from one longitudinal side, an extension direction of the first light channel and an extension direction of the second light channel are different from each other.
9. The razor assembly of claim 1, wherein the razor cartridge is pivotable about a first pivot axis parallel to a longitudinal direction with respect to the razor handle.
10. The razor assembly of claim 9, wherein, when the position at which the razor cartridge does not pivot about the first pivot axis is referred to as a first neutral position, the amount of light emitted to the outside at the cartridge top surface is greatest in the first neutral position.
11. The razor assembly of claim 9, wherein the first pivot axis is disposed between the light source unit and the at least one razor blade.
12. The razor assembly of claim 9, wherein the light source unit comprises: a first circuit board comprising a support surface; and a light emitting unit disposed on the support surface and electrically connected to the first circuit board, wherein, when the position at which the razor cartridge does not pivot about the first pivot axis is referred to as a first neutral position, the first circuit board is perpendicular to the cartridge top surface in the first neutral position.
13. The razor assembly of claim 1, wherein the razor handle further comprises: a grip portion comprising a second circuit board that is electrically connected to a power supply unit; a connecting portion configured to electrically connect the second circuit board and the light source unit; and a head portion configured to accommodate at least a portion of the connecting portion, one side of which is connected to the grip portion, and the other side of which is connected to the razor cartridge, wherein the light source unit is positioned on one side of the head portion.
14. The razor assembly of claim 13, wherein the head portion comprises: a connecting head on one side of which is connected to the razor cartridge; and a head adapter connected to one side of the grip portion and configured to surround at least a portion of the connecting head, wherein the connecting head is pivotable about a second pivot axis in a direction perpendicular to a longitudinal direction with respect to the head adapter.
15. The razor assembly of claim 14, wherein: the head adapter and the connecting head are fastened by a fastening member; and the fastening member penetrates the head adapter, the connecting portion, and the connecting head.

16. The razor assembly of claim 14, wherein at least a portion of the light source unit is supported by the connecting head.
17. The razor assembly of claim 14, wherein: at least a portion of the connecting portion comprises a first bending area that is convex downward when viewed from one longitudinal side, when the position at which the connecting head does not pivot about the second pivot axis is referred to as a second neutral position, at least a portion of the connecting head is positioned at an upper portion of the first bending area with respect to a direction parallel to the second pivot axis in the second neutral position.
18. The razor assembly of claim 14, wherein: at least a portion of the connecting portion comprises a second bending area that is convex upward when viewed from one longitudinal side; and at least a portion of the connecting head supports the connecting portion at a lower portion of the second bending area with respect to a direction parallel to the second pivot axis.